## Type MS · Design validation checks for a code format

### Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part a in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (a) | A1<br>A2<br>A3<br>A4 | 2.1.4 | Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g.<br><br>First 3 characters are upper case letters: Fff123456o<br>Characters 4 to 9 are non-zero numbers: FFF101111o<br>Consists of only 10 characters: FFF111111ee<br>Final character correct for sum of numbers: FFF112233o | | (4) |

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Design validation checks for a code format

### Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part a in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (b) (i) | B1<br>B2 | 1.1.9 | Award 1 mark each for any of:<br>• Simple implementation (1)<br>• Can be used for sorted or unsorted lists (1)<br>• If the target is at the beginning of the data structure the search will be faster than a binary search | Accept disadvantages of a binary compared with a linear search. | (2) |

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part a in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (b) (ii) | B3<br>B4 | 1.1.9 | Award 1 mark each for any of:<br>• May need to compare with all items in list before search complete (1)<br>• May need longer time for searching a large list (1) | Accept advantages of a binary compared with a linear search. | (2) |

## Type MS · Design validation checks for a code format

### Answer 1 of 3 · Question 5 part a · 4 marks

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| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Design validation checks for a code format

### Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part a in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (c) (i) | C1<br>C2 | 2.6.2 | Award 1 mark each up to a maximum of 2 for:<br>• Both can use arguments / parameter passing (1)<br>• Both can use local variables (1)<br>• Both can be called from anywhere within the program (1)<br>• Code can be reused without being rewritten (1)<br>• Both can be library files (1)<br>• The code can be independently tested for both (1) | | (2) |

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part a in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (c) (ii) | C3<br>C4 | 2.6.2 | Award two marks for a linked explanation such as:<br><br>• A function must always return a result (1) whereas a procedure does not (1)<br>• A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1)<br>• The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1)<br>• A function produces information (1) whereas a procedure performs a task (1) | | (2) |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · Design validation checks for a code format

### Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part a in this mark scheme extract

**Mark Scheme – Python Coding**

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 1 of 10 · find part g in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | **The only correct answer is C**<br><br>A is not correct because Boolean only has values True and False<br>B is not correct because it only allows for one character<br>D is not correct because it is used for mixed text and numeric data | | (1) |

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 2 of 10 · find part g in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(b) | (an error that) produces incorrect /unintended output (1) | Accept an example of an explained logic error | (1) |

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 3 of 10 · find part g in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(d) | Any one from:<br>1. explain how the program works (1)<br>2. improve readability of the program (1)<br>3. make code more understandable (1) | | (1) |

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 4 of 10 · find part g in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(e) | 1. Global variables created in the main program / can be accessed from any part of the program / are in scope throughout the program (1)<br>2. Local variables are declared/created inside a function/blocked code (indented, brackets) and are only accessible there / are only in scope within the function in which they're declared (1) | | (2) |

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 5 of 10 · find part g in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

# Type MS · Design validation checks for a code format

## Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 6 of 10 · find part g in this mark scheme extract

<table>
<tr><td>1(g)</td><td>For up to two tests<br>Award 1 mark for an appropriate validation test and 1 mark for an item of test data that would fail only the given test<br><br><table><tr><th>Validation test</th><th>Erroneous data</th></tr><tr><td>Are the first two characters letters/upper case letters?</td><td>1234CCC</td></tr><tr><td>Are the third- and fourth-characters numerical digits?</td><td>CCCCBBB</td></tr><tr><td>Are the last three characters letters/upper case letters?</td><td>CC12345</td></tr></table></td><td>Each test should target a different feature of one of the requirements<ul><li>Exactly 7 characters (in the stem)</li><li>First two characters are letters / upper case letters</li><li>Characters 3 & 4 are numeric</li><li>Last three characters are letters / upper case letters</li></ul></td><td>(4)</td></tr>
</table>

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 7 of 10 · find part g in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(c) | **C#**<br>1. Semicolon at the end of line 9 [maxValue=5;] (1)<br>2. Consistent capitalisation of 'count' on lines 10, 11, 13 and 14 (1)<br>3. constantValue needs to be defined and given the value 7 / replaced by value 7 (1)<br>**Java**<br>1. Semicolon at the end of line 8 [maxValue = 5;] (1)<br>2. Consistent capitalisation of 'count' on lines 7, 9, 11 and 12 (1)<br>3. constantValue needs to be defined and given the value 7 / replaced by 7 (1)<br>**Python**<br>1. Colon needed at end of line 7 [count, maxValue):] (1)<br>2. Consistent capitalisation of 'count' on line 3, 7, 8 and 9 (1)<br>3. constantValue needs to be defined and given the value 7 / replaced by 7 (1) | Candidates are required to open the file Q01c in the code editor.<br>Amended code should be saved as Q01cFINISHED.<br>Line numbers may have been changed by corrections to code (e.g. insertion of constantValue definition) | (3) |

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 8 of 10 · find part g in this mark scheme extract

<table>
<tr>
<td>1(f)</td>
<td>Award 1 mark for each correct pair of relational operators and 1 mark for correct Boolean operator for each condition.<br><br>
<table>
<tr><th>Condition</th><th>Coding</th></tr>
<tr><td rowspan="2">Shop income more than £5000 or assistant sold at least 10 pairs</td><td>shopIncome > 5000; assistantSales >= 10 (1)</td></tr>
<tr><td>correct OR operator (1)</td></tr>
<tr><td rowspan="2">Shop income at least £2000 and assistant sold at least 5 pairs</td><td>shopIncome >= 2000; assistantSales >= 5 (1)</td></tr>
<tr><td>correct AND operator (1)</td></tr>
</table>
<br>Accept >9, >1999 and >4 for respective >= comparisons</td>
<td>Candidates are required to open the file Q01f in the code editor.<br>Amended code should be saved as Q01fFINISHED.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Do not award Boolean operator mark for single | or & as these are not correct.</td>
<td>(4)</td>
</tr>
</table>

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 9 of 10 · find part g in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Design validation checks for a code format

### Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 10 of 10 · find part g in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

# Type MS · Design validation checks for a code format

## Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2(a) | A1 | **The only correct answer is D**<br><br>***A*** *is not correct because abstraction is a technique used in computational thinking*<br>***B*** *is not correct because encryption is a security measure*<br>***C*** *is not correct because iteration is a programming construct* | | **1** |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 2(b)(i) | | Award **one** mark for each correct cell: | | |
| | B1<br>B2 | real/double/float: (1)<br>• pay rate/18.75 (1)<br>• total pay/356.25 (1)<br>• tax/0.2/71.25 (1)<br>• income/285.00 (1)<br><br>int/integer: (1)<br>• hours/19 (1) | Example must match data type for B2<br><br>Award suitable example even if not taken from payslip<br><br>Accept string and matching example e.g. Hours | **2** |

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# Type MS · Design validation checks for a code format

## Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 2(b)(ii) | | Award **one** mark for each correct cell: | | |
| | B3 | **Input:**<br>• Pay Rate/18.75 (1)<br>• Hours/19 (1) | | |
| | B4 | **Process**:<br>• Pay Rate * Hours/18.75 * 19 (1)<br>• Total Pay * Tax/356.25 * 0.2 (1)<br>• Total Pay – Tax/356.25 – 71.25 (1) | Don't need output e.g. total pay, just the calculation part e.g. pay rate * hours:<br>Total pay = pay rate * hours<br>Accept 0.2 or 20% as part of calculation for tax<br>Ok to mix types e.g. Total Pay * 20%<br>Accept descriptions such as *multiply the pay rate by hours (hours etc could be taken from input box/ B3 mark)* | |
| | B5 | **Output:**<br>• Total Pay/356.25 (1)<br>• Tax/71.25 (1)<br>• Income/285 (1) | | **3** |

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# Type MS · Design validation checks for a code format

## Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 2(c)(i) | C1<br>C2 | Award **one** mark for **one** correct cell:<br>Award **two** marks for all **three** correct cells:<br>• **Normal:** any value between 0-255 e.g. 125<br>• **Boundary:** 0/255<br>• **Erroneous:** any negative value/any value 256 or above | Accept -1/256 for boundary data<br>Don't accept non-int types | **2** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 2(c)(ii) | | Award **one** mark for any of the following up to a maximum of **four** marks: | | |
| | C3 | Input changed to int before comparison (1) | | **4** |
| | C4 | Relational operator used correctly to check for valid input (both **>=**0 and **<**256 or equivalent) (1) | Allow:<br>>0 and/or <255 | |
| | C5 | 'and' logical operator used to combine relational tests (1) | or operator acceptable if candidate rearranges the selection statement and test works for correct code option – check if C4 is now awardable | |
| | C6 | Calling of `getHex(userInput)` (1) | Call must be inside print ( )<br>Subprogram and argument must be functionally correct<br>Ignore additional variables/strings in print statement | |

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## Type MS · Design validation checks for a code format

### Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
37 # complete the line to convert the user input to integer
38 userInput = int(input("please enter a number between 0 and 255 \n"))
39
40 # complete the test to check the input is between 0 and 255
41 if (userInput >=0) and (userInput <256):
42     # complete the line to call getHex subprogram
43     print(getHex(userInput))
44 else:
45     print("number out of range")
```

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## Type MS · Design validation checks for a code format

### Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|